

Solution

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

Second Semester, Session: 2025 – 2026

Control Systems (ECE/EEE/INSTR F242)

MM: 06

Time: 12 Minutes

Quiz-2

Date 10.02.2026

Note: Write your final answers in the space provided only. Show all the necessary steps in solution.

ID: Solution

Name: solution

Section:

1. For the signal flow graph shown in Fig. Q1, find the transfer function C/R .

[4]

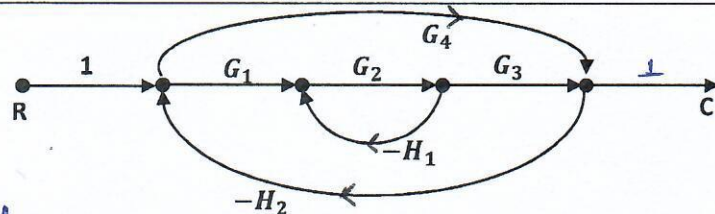


Fig. Q1

forward Paths,

$P_1 = G_1 G_2 G_3$; $P_2 = G_4$

Loops;

$L_1 = -G_2 H_1$; $L_2 = -G_1 G_2 G_3 H_2$; $L_3 = -G_4 H_2$

$\Delta = 1 - (L_1 + L_2 + L_3) + (L_1 L_3)$

$= 1 + G_2 H_1 + G_1 G_2 G_3 H_2 + G_4 H_2 + G_2 G_4 H_1 H_2$

$\Delta_1 = 1$; $\Delta_2 = (1 + G_2 H_1)$

$\frac{C}{R} = \frac{G_1 G_2 G_3 + G_4 (1 + G_2 H_1)}{1 + G_2 H_1 + G_1 G_2 G_3 H_2 + G_4 H_2 + G_2 G_4 H_1 H_2}$

$\frac{C}{R} = \frac{G_1 G_2 G_3 + G_4 (1 + G_2 H_1)}{(1 + G_2 H_1)(1 + G_4 H_2) + G_1 G_2 G_3 H_2}$

2. For a transfer function, $T(s) = \alpha / (s^2 + 2s + \alpha)$, find the sensitivity S_α^T for $\alpha = 2$ at $\omega = 0.1$ rad/s.

[2]

$S_\alpha^T = \frac{\partial T}{\partial \alpha} \times \frac{\alpha}{T} = \frac{(s^2 + 2s + \alpha) - (\alpha)}{(s^2 + 2s + \alpha)^2} \times \frac{\alpha}{\alpha / (s^2 + 2s + \alpha)}$

$S_\alpha^T = \frac{s(s+2)}{(s^2 + 2s + \alpha)}$

$S_\alpha^T \Big|_{\substack{\omega=0.1 \\ \alpha=2}} = \frac{j\omega(j\omega+2)}{(-\omega^2 + 2j\omega + 2)} \Big|_{\omega=0.1}$

$= \frac{\omega \sqrt{4 + \omega^2}}{\sqrt{(2 - \omega^2)^2 + 4\omega^2}} \Big|_{\omega=0.1} \approx 0.1$